

SESSION 3 • LECTURE 3

Displacement Work and the p-V Diagram

Detailed lecture notes — derivation of $\int p \, dV$, process work formulas with proofs, solved numericals

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Learning objectives. After this session you should be able to: (i) state the sign convention for heat and work; (ii) derive $\delta W = p \, dV$ for a quasi-static process; (iii) interpret work as area on the p-V diagram; (iv) compute work for constant-volume, constant-pressure, isothermal and polytropic processes; (v) find the polytropic index from two end states; (vi) recognise where $\int p \, dV$ fails.

1. Work Transfer and Sign Convention

In thermodynamics, **work is done by a system** on its surroundings if the sole effect external to the system *could be reduced to the raising of a weight*. This definition covers shaft rotation, electric current and boundary motion alike.

- **Sign convention (this course):** work done *by* the system is positive; work done *on* the system is negative. Heat supplied to the system is positive; heat rejected is negative.
- Units: work and heat are energy in transit — joule (J), kilojoule (kJ). Rate of work = power (W, kW).
- Work and heat are boundary phenomena and path functions: δW , δQ are inexact differentials.

2. Derivation of Displacement (pdV) Work

Consider gas at pressure p in a cylinder closed by a piston of area A , undergoing a quasi-static expansion.

1. Force exerted by the gas on the piston: $F = pA$.
2. Let the piston move outward through an infinitesimal distance dL .
3. Work done by the gas: $\delta W = F \cdot dL = pA \cdot dL$.
4. But $A \cdot dL = dV$, the increase in gas volume. Hence **$\delta W = p \, dV$** .
5. For a finite quasi-static process 1→2: **$W_{1-2} = \int_1^2 p \, dV$** .

$$\delta W = p \, dV \quad W_{1-2} = \int_1^2 p \, dV$$

Valid ONLY for quasi-static (fully resisted) processes

Graphical meaning. On the p-V diagram, $p \, dV$ is the area of an elementary vertical strip beneath the curve; the integral is the total **area under the process curve** between V_1 and V_2 . Since different curves joining the same end states enclose different areas, work depends on the path — the analytical proof that work is a path function.

3. Work Done in Standard Processes

Process	p-V relation	$W_{1-2} = \int p \, dV$	Notes
Constant volume (isochoric)	$V = C$	$W = 0$	$dV = 0$; heating a rigid tank
Constant pressure (isobaric)	$p = C$	$W = p(V_2 - V_1)$	Rectangle of area $p \times \Delta V$
Isothermal (ideal gas)	$pV = C$	$W = p_1 V_1 \ln(V_2/V_1) = mRT \ln(V_2/V_1)$	Also = $mRT \ln(p_1/p_2)$
Polytropic	$pV^n = C \, (n \neq 1)$	$W = (p_1 V_1 - p_2 V_2)/(n - 1)$	General compression/expansion
Reversible adiabatic	$pV^\gamma = C$	$W = (p_1 V_1 - p_2 V_2)/(\gamma - 1)$	$\gamma = c_p/c_v$ (≈ 1.4 for air)

Derivation of the isothermal result

For $pV = p_1 V_1 = C$: $W = \int p \, dV = \int (p_1 V_1/V) \, dV = p_1 V_1 \int dV/V = p_1 V_1 \ln(V_2/V_1)$. Using $pV = mRT$, this equals $mRT \ln(V_2/V_1)$.

Derivation of the polytropic result

For $pV^n = C$: $W = \int C V^{-n} \, dV = C [V^{1-n}/(1-n)]_1^2 = (p_2 V_2 - p_1 V_1)/(1-n) = (p_1 V_1 - p_2 V_2)/(n-1)$. Setting $n = \gamma$ gives the adiabatic case; $n \rightarrow 1$ recovers the logarithmic (isothermal) form.

Finding the index n from two states

$$n = \ln(p_1/p_2) / \ln(V_2/V_1)$$

Take logs of $p_1 V_1^n = p_2 V_2^n$

4. Worked Examples

WORKED EXAMPLE 3.1 — Polytropic index and work from two end states

A gas expands from 5 bar, 0.1 m³ to 1 bar, 0.4 m³ following $pV^n = C$. Determine n and the work done.

Solution.

1. $n = \ln(p_1/p_2)/\ln(V_2/V_1) = \ln 5 / \ln 4 = 1.609/1.386 = 1.161$.
2. Convert: $p_1 = 500$ kPa, $p_2 = 100$ kPa.
3. $W = (p_1 V_1 - p_2 V_2)/(n-1) = (500 \times 0.1 - 100 \times 0.4)/0.161 = (50 - 40)/0.161 = 62.1$ kJ.
4. Positive $\Rightarrow$ expansion work done by the gas.

ANSWER $n \approx 1.16$ | $W \approx +62.1$ kJ

WORKED EXAMPLE 3.2 — Isothermal compression

1.2 kg of air ($R = 0.287$ kJ/kg·K) at 350 K is compressed isothermally from 1 bar to 4 bar. Find the work done and comment on its sign.

Solution.

1. Isothermal: $W = mRT \ln(p_1/p_2)$ (volume ratio = inverse pressure ratio).
2. $W = 1.2 \times 0.287 \times 350 \times \ln(1/4) = 120.54 \times (-1.386) = -167.1$ kJ.
3. Negative $\Rightarrow$ work is done ON the air, as expected for compression.

ANSWER $W \approx -167$ kJ

Note: $\ln(1/4) = -\ln 4$; the sign emerges automatically from the ratio.

WORKED EXAMPLE 3.3 — Work for a three-process path

A gas at 1 bar, 0.10 m³ undergoes: (i) constant-pressure heating to 0.30 m³; (ii) constant-volume pressure rise to 3 bar; (iii) isothermal-shaped return is NOT taken — instead the gas stays at state 3. Find the total work for (i)+(ii) and sketch the path.

Solution.

1. Step (i): $W_1 = p\Delta V = 100 \times (0.30 - 0.10) = 20$ kJ.
2. Step (ii): $W_2 = 0$ (rigid vertical line on p-V, $dV = 0$).
3. Total $W = 20 + 0 = 20$ kJ; the p-V sketch is a horizontal line followed by a vertical rise.

ANSWER $W_{\text{total}} = +20$ kJ

Note: Work accumulates only along segments where volume changes.

5. When $\int p \, dV$ Fails — and Other Work Modes

- **Free (unrestrained) expansion:** gas expanding into a vacuum meets no resisting pressure, so $W = 0$ although V increases. The process is not quasi-static; $\int p \, dV$ must not be used.
- **Rapid processes:** pressure is non-uniform; the integral is undefined.
- **Other quasi-static work modes:** shaft/paddle work, electrical work ($\int VI \, dt$), spring work ($\frac{1}{2}k(x_2^2 - x_1^2)$), surface-tension and elastic-wire work. All obey the same sign convention, and the net work is the algebraic sum of every mode crossing the boundary.

KEY POINT Before applying any formula ask: is the process quasi-static, and is the gas fully resisted at the moving boundary? If not, evaluate work from its definition, not from $\int p \, dV$.

6. Practice Problems (answers given)

#	Problem	Answer
P1	Gas expands at constant pressure 2.5 bar from 0.08 m ³ to 0.20 m ³ . Find W .	+30 kJ
P2	Air at 6 bar, 0.03 m ³ expands to 0.12 m ³ with $pV^{1.35} = C$. Find p_2 and W .	$p_2 \approx 0.925$ bar; $W \approx +19.7$

		kj
P3	Show that for the same volume ratio, isothermal expansion of an ideal gas delivers more work than adiabatic expansion. (Hint: compare the curves' steepness / areas.)	Isothermal curve lies above adiabatic $\Rightarrow$ larger area
P4	0.8 kg of air at 300 K doubles its volume isothermally. Find W.	+47.8 kj

7. Summary

- W by system positive, Q to system positive; both are path functions.
- $\delta W = p dV$; W = area under the quasi-static p-V curve.
- Memorise the five work formulas and their p-V relations; recover n from $n = \ln(p_1/p_2)/\ln(V_2/V_1)$.
- Free expansion does zero work; $\int p dV$ applies only to fully resisted quasi-static processes.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 3 (Work and Heat Transfer). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.